
系统开发工具基础

实验报告

实验周次：第 1 周

学 院 信息科学与工程学部

专 业 计算机科学与技术

姓 名 臧文商

学 号 25020007160

实验日期 2026.8.24

一、 课堂练习

1. 题目一：含空格文件名的批量整理

(1) 题目内容

题目要求

- 创建 input/docs 和 input/tmp；创建 input/docs/notes one.txt（内容为 alpha 和 beta 两行）、input/docs/.secret.txt（内容为 hidden）、input/tmp/empty.txt（空文件）以及 input/run.log（任意两行日志）。
- 显示 q01 的绝对路径，并以长格式列出 input 下全部项目，包含隐藏文件。
- 把 input 下所有.txt 文件复制到 work/<学号>/，保留相对目录结构并正确处理名称中的空格。
- 将复制后的目录权限设为 750、普通文件权限设为 640；生成 inventory.txt，列出相对路径和字节数。

(2) 解题过程

按任务要求输入指令如图

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ mkdir -p input/docs input/tmp

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ echo -e "alpha\nbeta" > "input/docs/notes one.txt"

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ echo "hidden" > input/docs/.secret.txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ touch input/tmp/empty.txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ echo -e "加油华为\n加油CHina" > input/run.log

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ pwd
/c/Users/guolicheng/Desktop/系统开发工具基础/实验/week1/q01

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ ls -la input/
total 1
drwxr-xr-x 1 guolicheng 197121 0 Aug 29 17:47 ./
drwxr-xr-x 1 guolicheng 197121 0 Aug 29 17:42 ../
drwxr-xr-x 1 guolicheng 197121 0 Aug 29 17:44 docs/
-rw-r--r-- 1 guolicheng 197121 25 Aug 29 17:47 run.log
drwxr-xr-x 1 guolicheng 197121 0 Aug 29 17:45 tmp/

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ mkdir -p work/25020007160

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ find input -name "*.txt" -exec cp --parents {} work/25020007160/ \;
```

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ find work/25020007160 -type d -exec chmod 750 {} \;

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ find work/25020007160 -type f -exec chmod 640 {} \;

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ cd work/25020007160

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01/work/25020007160 (main)
$ find work/25020007160 -type f -printf "%p %s\n" > inventory.txt
find: 'work/25020007160': No such file or directory

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01/work/25020007160 (main)
$ find . -type f -printf "%p %s\n" > inventory.txt
```

(3) 实验结果

得到目录结构和 inventory.txt 内容如下，符合实验预期。

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ find .
./input
./input/docs
./input/docs/.secret.txt
./input/docs/notes one.txt
./input/run.log
./input/tmp
./input/tmp/empty.txt
./work
./work/25020007160
./work/25020007160/input
./work/25020007160/input/docs
./work/25020007160/input/docs/.secret.txt
./work/25020007160/input/docs/notes one.txt
./work/25020007160/input/tmp
./work/25020007160/input/tmp/empty.txt
./work/25020007160/inventory.txt
```

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01/work/25020007160 (main)
$ cat inventory.txt
./input/docs/.secret.txt 7
./input/docs/notes one.txt 11
./input/tmp/empty.txt 0
./inventory.txt 0
```

2. 题目二：随机访问日志统计

(1) 题目内容

在 q02 中创建 access.csv，并使用下方给定数据完成日志统计。

给定内容

```
timestamp,user,path,status,latency_ms
09:00:01,alice,/api/users,200,120
09:00:02,bob,/api/orders,500,310
09:00:03,alice,/api/users,503,180
09:00:04,carol,/login,500,90
09:00:05,bob,/api/orders,502,260
09:00:06,dave,/health,200,20
09:00:07,alice,/api/orders,500,420
09:00:08,carol,/api/users,500,150
```

题目要求

- 编写 analyze.sh，接收一个 CSV 文件路径；文件不存在时将错误写入 stderr 并返回非零退出码。
- 使用 Shell 管道以及 awk、sort、head 等工具，输出 HTTP 5xx 数量最多的前 2 个 path；按次数降序，次数相同时按 path 字典序。
- 输出全部数据行的平均 latency_ms，保留两位小数；表头不得计入。
- 分别对 access.csv 和一个不存在的文件运行脚本；不得使用 Python、电子表格或手工统计。

(2) 解题过程

按要求创建 access.csv，编写 access.sh 过程如下

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1 (main)
$ mkdir q02

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1 (main)
$ cd q02

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q02 (main)
$ vi access.csv

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q02 (main)
$ vi analyze.sh

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q02 (main)
$ chmod +x analyze.sh
```

```
# 脚本接收1个参数: csv文件路径
if [ $# -ne 1 ]
then
    echo "使用方式: $0 输入csv文件" >&2
    exit 1
fi

infile="$1"

# 判断普通文件是否存在
if [ ! -f "$infile" ]
then
    echo "错误: 文件 $infile 不存在" >&2
    exit 2
fi

echo "HTTP 5xx最多的前2个path:"
awk -F ',' 'NR>1 && $4 ~ /\^5[0-9][0-9]$/ {count[$3] = count[$3] + 1}
END {
    for(p in count) {
        print count[p], p
    }
}' "$infile" | sort -k1,1nr -k2,2 | head -2

echo ""
echo "latency_ms平均值:"
awk -F ',' 'NR>1 {total = total + $5; num = num + 1}
END {
    printf "%.2f\n", total / num
}' "$infile"
```

(3) 实验结果

分别对 access.csv 和一个不存在的文件运行脚本，结果符合预期

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q02 (main)
$ ./analyze.sh access.csv
HTTP 5xx最多的前2个path:
3 /api/orders
2 /api/users

latency_ms平均值:
172.22

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q02 (main)
$ ./analyze.sh 123.csv
错误: 文件 123.csv 不存在

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q02 (main)
$ echp $?
bash: echp: command not found

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q02 (main)
$ echo $?
127
```

3. 题目三：制造、解决并解释一次合并冲突

(1) 题目内容

在 q03 中从零创建一个 Git 仓库，并主动制造、解决一次合并冲突。

题目要求

- 初始化仓库，在 main 创建 config.txt，内容为 mode=normal，并完成初始提交。
- 创建 feature-a，把内容改为 mode=safe 并提交；从初始 main 创建 feature-b，把内容改为 mode=fast 并提交。
- 回到 main，先合并 feature-a，再合并 feature-b；解决冲突时保留 mode=safe，并另加一行 note=reviewed。
- 确认工作区干净，运行 git log --all --graph --decorate --oneline 查看历史。

(2) 解题过程

按题目要求初始化仓库和创建分支如下

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (main)
$ git init
Initialized empty Git repository in c:/Users/guolicheng/Desktop/系统开发工具基础/实验/week1/q03/.git/

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master)
$ echo "mode=normal" > "config.txt"

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master)
$ git add config.txt
warning: in the working copy of 'config.txt', LF will be replaced by CRLF the next time Git touches it

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master)
$ git commit -m "mode=normal"
[master (root-commit) 85d1dc6] mode=normal
1 file changed, 1 insertion(+)
 create mode 100644 config.txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master)
$ git checkout -b feature-a
Switched to a new branch 'feature-a'

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-a)
$ echo "mode=safe" > config.txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-a)
$ git add config.txt
warning: in the working copy of 'config.txt', LF will be replaced by CRLF the next time Git touches it

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-a)
$ git commit -m "mode=safe"
[feature-a 64ed774] mode=safe
1 file changed, 1 insertion(+), 1 deletion(-)

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-a)
$ git checkout main
error: pathspec 'main' did not match any file(s) known to git

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-a)
$ git checkout master
Switched to branch 'master'

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master)
$ git checkout -b feature-b
Switched to a new branch 'feature-b'

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-b)
$ echo "mode=fast" > config.txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-b)
$ git add config.txt
warning: in the working copy of 'config.txt', LF will be replaced by CRLF the next time Git touches it

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-b)
$ git commit -m "mode=fast"
[feature-b 2efe8dc] mode=fast
1 file changed, 1 insertion(+), 1 deletion(-)
```

合并并产生冲突如下

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-a)
$ git checkout main
error: pathspec 'main' did not match any file(s) known to git

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-a)
$ git checkout master
Switched to branch 'master'

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master)
$ git checkout -b feature-b
Switched to a new branch 'feature-b'

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-b)
$ echo "mode=fast" > config.txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-b)
$ git add config.txt
warning: in the working copy of 'config.txt', LF will be replaced by CRLF the next time Git touches it

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-b)
$ git commit -m "mode=fast"
[feature-b 2efe8dc] mode=fast
1 file changed, 1 insertion(+), 1 deletion(-)

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-b)
$ git checkout master
Switched to branch 'master'

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master)
$ git merge feature-a -m "合并feature-a"
Updating 85d1dc6..64ed774
Fast-forward (no commit created; -m option ignored)
 config.txt | 2 +-
1 file changed, 1 insertion(+), 1 deletion(-)

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master)
$ git merge feature-b -m "合并feature-b"
Auto-merging config.txt
CONFLICT (content): Merge conflict in config.txt
Automatic merge failed; fix conflicts and then commit the result.
```

进入 config.txt 解决冲突如下

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master|MERGING)
$ vi config.txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master|MERGING)
$ git add config.txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master|MERGING)
$ git commit -m "解决冲突"
[master f10f4a2] 解决冲突
```

```
mode=safe
note=reviewed
```

(3) 实验结果

确认工作区干净后，查看历史如图，符合要求

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master)
$ git status
On branch master
nothing to commit, working tree clean

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master)
$ git log --all --graph --decorate --oneline
* f10f4a2 (HEAD -> master) 解决冲突
|
| * 2efe8dc (feature-b) mode=fast
| * | 64ed774 (feature-a) mode=safe
|/
* 85d1dc6 mode=normal
```

4. 题目四：修复并构建一页技术说明

(1) 题目内容

将下方模板保存为 q04/report.tex，补全内容并从命令行构建 PDF。

给定内容

```
\documentclass{article}
\usepackage{amsmath}
\begin{document}
\title{Tool Report}
\author{姓名-学号}
\maketitle
\section{Result}
% 在此加入公式、表格及正文引用
\end{document}
```

题目要求

- 加入一个带编号和 label 的公式，并在正文中使用 ref 或 eqref 引用。
- 加入一个 2 行 2 列表格，设置 caption 和 label，并在正文中引用该表。
- 使用 latexmk -pdf -halt-on-error 或 tectonic 生成 report.pdf。
- 确认 PDF 能打开，且交叉引用不显示 “??” 。

(2) 解题过程

按要求修改 report.tex 如图并生成 pdf

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q04 (main)
$ vi report.tex

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工
具基础/实验/week1/q04 (main)
$ latexmk -pdf -halt-on-error report.tex
Initial win CP for (console input, console output, system): (CP936, CP936, CP936)
I changed them all to CP936
Rc files read:
  NONE
Latexmk: This is Latexmk, John Collins, 15 June 2025. ve
rsion 4.87.
No existing .aux file, so I'll make a simple one, and re
quire run of *latex.
```



```

\documentclass{article}
\usepackage{amsmath}
\begin{document}
\title{Tool Report}
\author{Zang Wenshang-25020007160}
\maketitle
\section{Result}

The arithmetic series summation is given by Eq.~\eqref{eq:series}.
\begin{equation}\label{eq:series}
\sum_{i=1}^n i = \frac{n(n+1)}{2}
\end{equation}

Experimental parameters are listed in Table~\ref{tab:param}.
\begin{table}[htbp]
\centering
\caption{Parameter Statistics}
\label{tab:param}
\begin{tabular}{|c|c|}
\hline
Sample Size & Valid Sample \\
\hline
200 & 186 \\
\hline
\end{tabular}
\end{table}

We can compute sequence sum using Eq.~\eqref{eq:series}. Table~\ref{tab:param} shows the experimental statistical parameters.
\end{document}

```

(3) 实验结果

经检验 pdf 符合预期，交叉引用未显示??

Tool Report

Zang Wenshang-25020007160

August 29, 2026

1 Result

The arithmetic series summation is given by Eq. (1).

$$\sum_{i=1}^n i = \frac{n(n+1)}{2} \quad (1)$$

Experimental parameters are listed in Table 1.

Table 1: Parameter Statistics

Sample Size	Valid Sample
200	186

We can compute sequence sum using Eq. (1). Table 1 shows the experimental statistical parameters.

二、 课后练习

1. 题目一

(1) 题目内容

ls 的 -l 选项 (flag) 作用是什么? 运行 `ls -l /` 并观察输出。每一行最前面的 10 个字符分别代表什么? (提示: `man ls`)

(2) 解题过程

使用 `ls -help` 查看 ls 的 -l 选项的作用, 运行 `ls -l /` 并输出

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/1 (main)
$ ls --help
Usage: ls [OPTION]... [FILE]...
List information about the FILES (the current directory by default).
Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.
```

```
-l          used only with -S and per-directory co...
            use a long listing format
            show basic file information for each
```

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/1 (main)
$ ls -l / > ls_root.txt
```

```
total 10832
-rw-r--r-- 1 guolicheng 197121  18765 Aug 21 00:03 LICENSE.txt
-rw-r--r-- 1 guolicheng 197121 312948 Aug 21 00:03 ReleaseNotes.html
drwxr-xr-x 1 guolicheng 197121    0 Aug 24 14:19 bin/
drwxr-xr-x 1 guolicheng 197121    0 Aug 24 14:19 cmd/
drwxr-xr-x 1 guolicheng 197121    0 Aug 24 14:19 dev/
drwxr-xr-x 1 guolicheng 197121    0 Aug 24 14:19 etc/
-rwxr-xr-x 1 guolicheng 197121 135016 Aug 20 23:47 git-bash.exe*
-rwxr-xr-x 1 guolicheng 197121 134488 Aug 20 23:47 git-cmd.exe*
drwxr-xr-x 1 guolicheng 197121    0 Aug 24 14:19 mingw64/
dr-xr-xr-x 9 guolicheng 197121    0 Aug 30 12:56 proc/
drwxr-xr-x 1 guolicheng 197121    0 Aug 30 12:47 tmp/
-rw-r--r-- 1 guolicheng 197121 1932106 Aug 24 14:19 unins001.dat
-rwxr-xr-x 1 guolicheng 197121 4449992 Aug 24 14:19 unins001.exe*
-rw-r--r-- 1 guolicheng 197121  25396 Aug 24 14:19 unins001.msg
drwxr-xr-x 1 guolicheng 197121    0 Aug 24 14:19 usr/
```

(3) 实验结果

经实验得到结论：-l 选项的作用是以长列表格式输出，显示文件类型、权限、硬链接数、所有者、用户组、文件大小、修改时间以及文件名。每行开头 10 个字符，第一位代表文件类型，d 是目录，- 是普通文件，l 是符号链接。第 2-4 位是文件所有者的读写执行权限，第 5-7 位是同组用户的读写执行权限，第 8-10 位是其他用户的读写执行权限。

2. 题目二

(1) 题目内容

Shell 有三条标准流：stdin、stdout、stderr。运行 `ls /nonexistent /tmp`，把 stdout 和 stderr 分别重定向到两个文件。你将如何把两者都重定向到同一个文件？

(2) 解题过程

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/2 (main)
$ ls /nonexistent /tmp > out.txt 2> err.txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/2 (main)
$ cat out.txt
/tmp:
000cgfdo.tbu/
005415bf-876c-4062-b520-684f1bebe024_Array.h.rar.024/
01fc14f.tmp@
02ob0d1b.fx2/
02q79CF.tmp@
```

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/2 (main)
$ cat err.txt
ls: cannot access '/nonexistent': No such file or directory

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/2 (main)
$ ls /nonexistent /tmp > all.txt 2>&1

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/2 (main)
$ cat all.txt
ls: cannot access '/nonexistent': No such file or directory
/tmp:
000cgfdo.tbu/
005415bf-876c-4062-b520-684f1bebe024_Array.h.rar.024/
01fc14f.tmp@
02ob0d1b.fx2/
02q79CF.tmp@
```

(3) 实验结果

执行 `ls /nonexistent /tmp > out.txt 2> err.txt`，标准输出保存到 out.txt，标准错误保存到 err.txt。使用 `ls /nonexistent /tmp > all.txt 2>&1`，把标准输出和标准错误输出全部写入同一个 all.txt 文件。

3. 题目三

(1) 题目内容

写一个脚本，接收文件名参数 (\$1)，用 `test -f` 或 `[ -f ... ]` 检查该文件是否存在，并根据结果输出不同提示。

(2) 解题过程

新建脚本并测试。

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/3 (main)
$ vi checkfile.sh

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/3 (main)
$ touch real.txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/3 (main)
$ ./checkfile.sh real.txt
文件 real.txt 存在

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/3 (main)
$ ./checkfile.sh 1232.txt
文件 1232.txt 不存在
```

```
#!/bin/bash
if test -f "$1"
then
    echo "文件 $1 存在"
else
    echo "文件 $1 不存在"
fi
```

(3) 实验结果

经测试，脚本能够检查文件是否存在并输出相应提示。

4. 题目四

(1) 题目内容

在脚本的 set 选项（flag）里加入 -x 会发生什么？写个简单脚本试试并观察输出。

(2) 解题过程

新建 settest.sh 脚本，脚本内添加 set -x，编写简单变量赋值与命令。运行脚本观察终端输出。

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/4 (main)
$ vi settest.sh

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/4 (main)
$ ./settest.sh
bash: ./settest.sh: No such file or directory

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/4 (main)
$ ./settest.sh
+ a=10
+ b=20
+ echo 10 20
10 20
+ ls .
settest.sh
```

```
#!/bin/bash
set -x
a=10
b=20
echo $a $b
ls .
```

(3) 实验结果

根据实验得到结论：`set -x` 是 shell 调试选项，开启后会在行首以 `+` 打印每条即将执行的真实命令。未开启仅显示运行结果，开启后同时输出执行过程与结果。

5. 题目五

(1) 题目内容

写一条命令，把文件复制为带当天日期的备份文件名（例如 `notes.txt` → `notes_2026-01-12.txt`）。

(2) 解题过程

根据提示写出命令

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/5 (main)
$ touch notes.txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/5 (main)
$ cp notes.txt notes_$(date +%Y-%m-%d).txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/5 (main)
$ ls
notes.txt  notes_2026-08-30.txt
```

(3) 实验结果

成功把文件复制为带当天日期的备份文件名（如上图）。

6. 题目六

(1) 题目内容

使用管道找出你「home 目录」中最常见的 5 种文件扩展名。（提示：组合 `find`、`grep` / `sed` / `awk`、`sort`、`uniq -c` 以及 `head`）

(2) 解题过程

根据提示写出命令

```

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/6 (main)
$ find ~ -type f | awk -F '.' '{if(NF>1)print $NF}' | sort | uniq -c | sort -nr | head -5

find: '/c/Users/guolicheng/AppData/Local/Microsoft/Windows/INetCache/Low/Content.IE5': Permission denied
find: '/c/Users/guolicheng/AppData/Local/Temp/EEJ4AB7.tmp.dir': Permission denied
find: '/c/Users/guolicheng/AppData/Local/Temp/LISF_C7C37.tmp': Permission denied
find: '/c/Users/guolicheng/AppData/Local/Temp/nsa85CF.tmp': Permission denied
find: '/c/Users/guolicheng/AppData/Local/Temp/nsa9343.tmp': Permission denied
find: '/c/Users/guolicheng/AppData/Local/Temp/nsh70D0.tmp': Permission denied
find: '/c/Users/guolicheng/AppData/Local/Temp/hsi8D67.tmp': Permission denied
find: '/c/Users/guolicheng/AppData/Local/Temp/nsk37D6.tmp': Permission denied
find: '/c/Users/guolicheng/AppData/Local/Temp/nsoA512.tmp': Permission denied
find: '/c/Users/guolicheng/AppData/Local/Temp/nsp2B6F.tmp': Permission denied
find: '/c/Users/guolicheng/AppData/Local/Temp/nsp906E.tmp': Permission denied
find: '/c/Users/guolicheng/AppData/Local/Temp/nsq937B.tmp': Permission denied
find: '/c/Users/guolicheng/AppData/Local/Temp/nsrAE9D.tmp': Permission denied
find: '/c/Users/guolicheng/AppData/Local/Temp/nssA834.tmp': Permission denied
find: '/c/Users/guolicheng/AppData/Local/Temp/WinsAT': Permission denied
find: '/c/Users/guolicheng/AppData/Local/Temp/~nsu1.tmp': Permission denied
53276 py
23703 pyc
22031 js
15134 ts
12689 map

```

(3) 实验结果

管道串联多个命令完成统计，得到 home 目录出现最多的 5 种文件后缀 py,pyc,js,ts 和 map。

7. 题目七

(1) 题目内容

试着添加一个文件到一个仓库，先提交一些文件，然后从历史中删除该文件（不仅仅是最新的提交）。

(2) 解题过程

创建本地 Git 仓库先后提交 123.txt 和 hello.txt，使用 git filter-branch 重写历史移除 hello.txt 文件。

```

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/7 (main)
$ git init
Initialized empty Git repository in C:/Users/guolicheng/Desktop/系统开发工具基础/实验/week1/课后练习/

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/7 (master)
$ echo 123 > 123.txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/7 (master)
$ git add 123.txt
warning: in the working copy of '123.txt', LF will be replaced by CRLF the next time Git touches it

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/7 (master)
$ git commit -m "提交123.txt"
[master (root-commit) 0e241de] 提交123.txt
1 file changed, 1 insertion(+)
create mode 100644 123.txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/7 (master)
$ echo hello > hello.txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/7 (master)
$ git add hello.txt
warning: in the working copy of 'hello.txt', LF will be replaced by CRLF the next time Git touches it

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/7 (master)
$ git commit -m "提交hello.txt"
[master c130ebc] 提交hello.txt
1 file changed, 1 insertion(+)
create mode 100644 hello.txt

```

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/7 (master)
$ git log --stat
commit c130ebccbbba3403e4ab6a3eb14a84085351196 (HEAD -> master)
Author: Youmoguolicheng <zangwenshang@qq.com>
Date: Sun Aug 30 19:30:06 2026 +0800

    提交hello.txt

hello.txt | 1 +
1 file changed, 1 insertion(+)

commit 0e241deb823b250a97fac8324766127715b1959a
Author: Youmoguolicheng <zangwenshang@qq.com>
Date: Sun Aug 30 19:29:15 2026 +0800

    提交123.txt

123.txt | 1 +
1 file changed, 1 insertion(+)

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/7 (master)
$ git filter-branch --force --index-filter "git rm --cached --ignore-unmatch hello.txt" HEAD
WARNING: git-filter-branch has a glut of gotchas generating mangled history
rewrites. Hit Ctrl-C before proceeding to abort, then use an
Proceeding with filter-branch...

Rewrite c130ebccbbba3403e4ab6a3eb14a84085351196 (2/2) (1 seconds passed, remaining 0 predicted)    rm 'hello.txt'

Ref 'refs/heads/master' was rewritten
```

(3) 实验结果

经验证，目标文件 `hello.txt` 从全部版本历史删除，并非仅删除最新提交。

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/7 (master)
$ git log --stat
commit 46d0871aabf813e5e389ede9dd7c52cd619d06de (HEAD -> master)
Author: Youmoguolicheng <zangwenshang@qq.com>
Date: Sun Aug 30 19:30:06 2026 +0800

    提交hello.txt

commit 0e241deb823b250a97fac8324766127715b1959a
Author: Youmoguolicheng <zangwenshang@qq.com>
Date: Sun Aug 30 19:29:15 2026 +0800

    提交123.txt

123.txt | 1 +
1 file changed, 1 insertion(+)

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/7 (master)
$ ls
123.txt
```

8. 题目八

(1) 题目内容

和许多命令行工具一样，Git 提供配置文件（点文件）名叫 `.gitconfig`。在该文件中创建别名，使得运行自定义简写命令，等价执行另一条 git 命令输出。可以直接编辑该文件，也可以用命令添加别名。

(2) 解题过程

使用 `git config` 配置别名，将 `git status` 简写为 `git st`，运行简写命令验证效果。

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/8 (main)
$ git config alias.st status

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/8 (main)
$ git st
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
```

(3) 实验结果

别名配置生效，git st 等价 git status。

9. 题目九

(1) 题目内容

实践

→ 撰写代码来生成下列公式：

$$E = mc^2 \quad (1)$$

$$\pi = \frac{c}{d} \quad (2)$$

$$\frac{d}{dx} e^x = e^x \quad (3)$$

$$\frac{d}{dx} \int_0^\infty f(s) ds = f(x) \quad (4)$$

$$f(x) = \sum_i 0^\infty \frac{f^{(i)}(0)}{i!} x^i \quad (5)$$

$$x = \sqrt{\frac{x_i}{z}} y \quad (6)$$

如果需要帮助，可以查看本页面的 [源代码](#)。

(2) 解题过程

编写 LaTeX 源文件录入六条带编号数学公式，使用 latexmk 编译生成 PDF。


```

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/9 (main)
$ vi formula.tex
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/9 (main)
$ latexmk -pdf -halt-on-error formula.tex
Initial win CP for (console input, console output, system): (CP936, CP936, CP936)
I changed them all to CP936
Rc files read:
  NONE
Latexmk: This is Latexmk, John Collins, 15 June 2025. Version 4.87.
No existing .aux file, so I'll make a simple one, and require run of *latex.
Latexmk: applying rule 'pdflatex'...
Rule 'pdflatex': Reasons for rerun
Category 'other':
  Rerun of 'pdflatex' forced or previously required:
    Reason or flag: 'Initial setup'

```

```

\documentclass{article}
\usepackage{amsmath}
\begin{document}
\begin{align}
e &= mc^2 \\
\pi &= \frac{c}{d} \\
\frac{d}{dx}e^x &= e^x \\
\frac{d}{dx}\int_0^\infty f(s)ds &= f(x) \\
f(x) &= \sum_{i=0}^\infty \frac{f^{(i)}(0)}{i!} x^i \\
x &= \sqrt{\frac{x_i}{z}}y
\end{align}
\end{document}
~
~
~

```

(3) 实验结果

编译成功，得到带有六条公式的 PDF。

文件 C:/Users/guolicheng/Desktop/系统开发工具基础/实验/week1/课后练习/9/formula.pdf

1 / 1

$$e = mc^2 \quad (1)$$

$$\pi = \frac{c}{d} \quad (2)$$

$$\frac{d}{dx}e^x = e^x \quad (3)$$

$$\frac{d}{dx} \int_0^\infty f(s)ds = f(x) \quad (4)$$

$$f(x) = \sum_{i=0}^\infty \frac{f^{(i)}(0)}{i!} x^i \quad (5)$$

$$x = \sqrt{\frac{x_i}{z}}y \quad (6)$$

10. 题目十

(1) 题目内容

实践 11

尝试画出下列表格：

Item	Quantity	Price(\$)
Nails	500	0.34
Wooden boards	100	4.00
Bricks	240	11.50

City	Year		
	2006	2007	2008
London	45789	46551	51298
Berlin	34549	32543	29870
Paris	49835	51009	51970

(2) 解题过程

编写 LaTeX 源文件，按样式绘制两个不同格式的表格，使用 latexmk 编译生成 PDF。

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/10 (main)
$ vi table.tex

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/10 (main)
$ latexmk -pdf -halt-on-error table.tex
Initial win CP for (console input, console output, system): (CP936, CP936, CP936)
I changed them all to CP936
Rc files read:
  NONE
Latexmk: This is Latexmk, John Collins, 15 June 2025. version 4.87.
No existing .aux file, so I'll make a simple one, and require run of *latex.
Latexmk: applying rule 'pdflatex'...
```

```

\documentclass{article}
\begin{document}

% 第一个表格
\begin{tabular}{l|r|r}
Item          & Quantity & Price(\$) \\
\hline
Nails         & 500      & 0.34 \\
Wooden boards & 100      & 4.00 \\
Bricks        & 240      & 11.50 \\
\end{tabular}

\vspace{1cm}

% 第二个表格
\begin{tabular}{l|rrr}
& \multicolumn{3}{c}{Year} \\
city & 2006 & 2007 & 2008 \\
\hline
London & 45789 & 46551 & 51298 \\
Berlin & 34549 & 32543 & 29870 \\
Paris  & 49835 & 51009 & 51970 \\
\end{tabular}

\end{document}
~

```

(3) 实验结果

编译成功，得到要求的表格。



Item	Quantity	Price(\$)
Nails	500	0.34
Wooden boards	100	4.00
Bricks	240	11.50

City	Year		
	2006	2007	2008
London	45789	46551	51298
Berlin	34549	32543	29870
Paris	49835	51009	51970

11. 题目十一

(1) 题目内容

设置全局 gitignore 文件，忽略操作系统和编辑器生成的临时文件（如.DS_Store、Thumbs.db、*.swp）。

(2) 解题过程

编辑全局 gitignore 文件，填入.DS_Store 、 Thumbs.db 、 *.swp 忽略规则，执行 git config 设置全局忽略路径，校验配置。

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/11 (main)
$ vi ~/.gitignore_global

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/11 (main)
$ git config --global core.excludesfile ~/.gitignore_global

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/课后练习/11 (main)
$ git config --global --get core.excludesfile
C:/Users/guolicheng/.gitignore_global
```

(3) 实验结果

全局 gitignore 配置生效，本机所有 Git 仓库自动忽略系统与编辑器产生的指定临时文件。

三、 解题感悟

通过本次实验，我系统掌握了 Shell、Git 与 LaTeX 等系统开发基础工具的核心用法。在命令行环境中，我实践了含空格文件名的批量处理、文件权限管理以及标准流重定向，并运用 find、awk 等工具实现日志统计与数据提取；在版本控制方面，我通过主动制造并解决 Git 合并冲突，深入理解了分支机制与冲突处理流程，同时掌握了历史重写与别名配置；在文档构建方面，我完成了带交叉引用的公式与表格排版，并成功编译 PDF。实验过程中，我不仅学会了多项新技能，还培养了严谨的工程实践思维，为后续系统开发奠定了坚实基础。

四、 GitHub 链接和 commit 记录截图

1. GitHub 链接

<https://github.com/Youmoguolicheng/system-dev-tool>

2. commit 记录截图

Commits on Aug 29, 2026

完成第四题 Youmoguolicheng committed yesterday	bff1230	 
完成第三题 Youmoguolicheng committed yesterday	b25d592	 
完成第二题 Youmoguolicheng committed yesterday	f960ad5	 
完成第一题 Youmoguolicheng committed yesterday	afcfef3	 
实验报告模板 Youmoguolicheng committed yesterday	ff35ad7	 

Commits on Aug 30, 2026

完成第十一题 Youmoguolicheng committed 6 minutes ago	bd6c800	 
完成课后实验第十题 Youmoguolicheng committed 17 minutes ago	b8ab588	 
完成课后第九题 Youmoguolicheng committed 42 minutes ago	9531b65	 
完成课后第八题 Youmoguolicheng committed 1 hour ago	0d51e75	 
完成课后第七题 Youmoguolicheng committed 1 hour ago	277dbde	 
完成课后第六题 Youmoguolicheng committed 2 hours ago	b78788d	 
完成课后第五题 Youmoguolicheng committed 2 hours ago	1937e90	 
完成第四题 Youmoguolicheng committed 7 hours ago	a5024d4	 
完成课后第三题 Youmoguolicheng committed 8 hours ago	341db81	 
完成课后第二题 Youmoguolicheng committed 8 hours ago	d3f009c	 
完成课后第一题 Youmoguolicheng committed 8 hours ago	b6b36b2	 
完成课堂实验部分的实验报告 Youmoguolicheng committed 10 hours ago	017cfb9	 

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1 (main)
$ git log --pretty=format:"%ad | %s" --date=iso
2026-08-30 21:20:17 +0800 | 完成第十一题
2026-08-30 21:09:50 +0800 | 完成课后实验第十题
2026-08-30 20:45:07 +0800 | 完成课后第九题
2026-08-30 20:16:45 +0800 | 完成课后第八题
2026-08-30 19:41:54 +0800 | 完成课后第七题
2026-08-30 19:14:13 +0800 | 完成课后第六题
2026-08-30 19:03:10 +0800 | 完成课后第五题
2026-08-30 13:41:58 +0800 | 完成第四题
2026-08-30 13:27:37 +0800 | 完成课后第三题
2026-08-30 13:11:16 +0800 | 完成课后第二题
2026-08-30 12:43:12 +0800 | 完成课后第一题
2026-08-30 11:11:44 +0800 | 完成课堂实验部分的实验报告
2026-08-29 21:40:39 +0800 | 完成第四题
2026-08-29 20:27:01 +0800 | 完成第三题
2026-08-29 19:45:43 +0800 | 完成第二题
2026-08-29 18:07:54 +0800 | 完成第一题
2026-08-29 16:19:15 +0800 | 实验报告模板
```